

MedForge Cheat Sheet — pituitary gland

Educational draft - source review required

Pituitary Gland – Cheat Sheet

Compact, evidence-based, student-friendly

Definition

Pea-sized endocrine gland at base of brain; regulates growth, metabolism, and hormone balance via hypothalamic control [S2].

Mechanism Chain

Hypothalamus → releases hormones (e.g., TRH, CRH, GnRH) → stimulates pituitary secretion → anterior lobe releases hormones (TSH, ACTH, GH, TSH, PRL); posterior lobe stores/releases ADH & oxytocin [S2].

Must-Know Facts

- Two lobes: anterior (adenohypophysis) and posterior (neurohypophysis) [S2].
- Anterior lobe secretes 5 hormones from 5 cell types [S2].
- Pituitary tumors are common (75–78 per 100,000) [S4].
- Symptoms: headaches, vision loss, nausea, endocrine dysfunction [S3].
- Hypersecretion causes: Cushing's syndrome, acromegaly, hyperprolactinemia [S4].
- Treatment: surgery (transsphenoidal) first-line for acromegaly/Cushing's; radiation/medications if needed [S4].
- Diagnosis: MRI, ACTH testing, physical exam [S4].

Red-Flag Confusions

- ☐ "Pituitary gland produces all hormones" → No; it's regulated by hypothalamus; only secretes specific hormones [S2].
- ☐ "Posterior lobe makes all hormones" → No; it stores and releases ADH/oxytocin from hypothalamus [S2].
- ☐ "All pituitary tumors are malignant" → False; most are benign; only 1–2% are cancerous [S3].

Quick Recall Prompts

- "Hypothalamus controls pituitary" → [S2]

- “Tumor symptoms: vision loss, headache” → [S3]
- “Cushing’s = ACTH overproduction” → [S4]
- “Surgery first for acromegaly” → [S4]
- “Anterior = 5 hormones” → [S2]

Sources: [S2], [S3], [S4]

Note: Evidence limited to provided references; no claims beyond cited data.

References

[S1] PubMed PMID 32634634 — Arterial Supply to the Pituitary Gland: A Comprehensive Review. <https://pubmed.ncbi.nlm.nih.gov/32634634/>

[S2] Physiology, Pituitary Gland - StatPearls - NCBI Bookshelf — [ncbi.nlm.nih.gov](https://www.ncbi.nlm.nih.gov/books/NBK459247/) (page retrieved 2026-09-14T14:34:34Z) <https://www.ncbi.nlm.nih.gov/books/NBK459247/>

[S3] Pituitary Tumor | Pituitary Gland - MedlinePlus — [medlineplus.gov](https://www.medlineplus.gov/pituitarytumors.html) (page retrieved 2026-09-14T14:34:38Z) <https://www.medlineplus.gov/pituitarytumors.html>

[S4] Pharmacoeconomic aspects of the treatment of pituitary gland ... — [pmc.ncbi.nlm.nih.gov](https://pmc.ncbi.nlm.nih.gov/articles/PMC3685378/) (page retrieved 2026-09-14T14:34:32Z) <https://pmc.ncbi.nlm.nih.gov/articles/PMC3685378/>

[S5] PubMed PMID 11130789 — The computed tomographic enhancement pattern of the normal canine pituitary gland. <https://pubmed.ncbi.nlm.nih.gov/11130789/>

[S6] PubMed PMID 11130789 — The computed tomographic enhancement pattern of the normal canine pituitary gland. <https://pubmed.ncbi.nlm.nih.gov/11130789/>